

KING'S LAND O'LAKES, WI, USA

WMO: 723123

Lat: 46.150N

Lon: 89.217W

Elev: 519

StdP: 95.24

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 14886

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -28.7	(c) -24.9	(d) -32.3	(e) 0.2	(f) -28.6	(g) -28.1	(h) 0.3	(i) -23.7	(j) 8.1	(k) -8.0	(l) 7.5	(m) -8.4	(n) 0.8	(o) 260	(p) 0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.1	(c) 29.7	(d) 20.0	(e) 27.9	(f) 18.9	(g) 26.5	(h) 18.1	(i) 21.8	(j) 26.6	(k) 20.8	(l) 24.9	(m) 20.0	(n) 24.1	(o) 3.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	16.1	23.7	19.5	15.2	22.8	18.6	14.3	22.0	66.4	26.7	62.8	25.1	59.5	23.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.7	(b) 6.9	(c) 5.8	(e) -33.4	(f) 32.4	(g) 2.0	(h) 1.4	(i) -34.9	(j) 33.3	(k) -36.1	(l) 34.1	(m) -37.2	(n) 34.9	(o) -38.7	(p) 35.9		
(a) 7.7	(b) 6.9	(c) 5.8	(e) -33.5	(f) 23.7	(g) 2.0	(h) 1.2	(i) -35.0	(j) 24.6	(k) -36.2	(l) 25.3	(m) -37.3	(n) 26.0	(o) -38.7	(p) 26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-10.6	-10.5	-3.4	3.3	10.9	15.9	18.2	16.7	12.9	6.2	-0.9	-7.7		
	DBStd	11.70	7.19	7.00	7.37	5.56	5.19	3.87	3.52	3.52	4.77	4.97	5.67	6.38		
	HDD10.0	2938	638	573	418	209	52	3	0	1	23	141	329	549		
	HDD18.3	5214	897	807	673	451	238	92	47	76	173	376	578	808		
	CDD10.0	871	0	0	4	8	78	179	256	208	111	25	1	0		
	CDD18.3	106	0	0	0	0	7	18	44	24	11	1	0	0		
	CDH23.3	1343	0	0	1	3	118	263	559	278	111	9	0	0		
	CDH26.7	290	0	0	0	0	28	53	141	50	18	0	0	0		
(14) Wind	WSAvg	2.4	2.6	2.6	2.6	2.9	2.7	2.3	2.0	1.9	2.1	2.5	2.6	2.4		
(15) Precipitation	PrecAvg	804	36	26	48	65	84	90	99	78	91	91	51	42		
	PrecMax	973	91	53	107	127	157	171	213	163	192	173	123	69		
	PrecMin	671	13	1	7	17	34	48	30	31	24	35	11	10		
	PrecStd	103	20	14	27	34	30	37	40	31	35	39	25	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	10.0	21.0	23.2	29.9	30.9	32.1	30.1	28.9	24.8	16.3	6.9		
		MCWB	2.6	5.2	14.0	13.3	17.8	20.9	21.7	20.1	20.2	15.5	11.0	4.3		
	2%	DB	2.7	4.8	14.5	19.6	26.9	28.0	30.0	27.9	26.3	21.2	12.7	3.9		
		MCWB	1.3	2.1	9.5	9.9	16.4	18.5	20.6	19.6	18.8	15.2	9.3	2.3		
	5%	DB	0.9	1.3	10.9	16.7	23.8	26.3	28.0	26.3	24.3	18.0	9.8	1.8		
		MCWB	0.3	-0.4	7.4	8.5	14.8	17.2	19.4	18.4	17.6	13.7	6.7	1.0		
	10%	DB	-0.3	-0.5	7.3	13.7	21.0	24.7	26.6	24.8	22.1	15.0	7.3	0.5		
		MCWB	-1.0	-1.8	3.5	7.3	13.3	16.4	18.6	17.8	16.7	11.2	5.0	0.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	5.7	14.9	14.7	20.4	22.5	24.1	22.5	21.7	20.0	12.7	5.8		
		MCDB	4.9	9.4	20.4	20.2	26.6	27.5	28.9	26.9	25.7	21.4	14.8	6.0		
	2%	WB	1.5	2.2	11.5	12.1	18.5	20.5	22.1	21.3	20.4	16.3	9.7	2.6		
		MCDB	2.6	4.0	13.2	17.0	22.7	25.5	27.3	25.1	23.8	19.8	12.3	3.4		
	5%	WB	0.2	-0.1	6.6	9.7	16.8	19.4	21.0	20.3	19.4	13.8	6.8	1.0		
		MCDB	0.8	1.4	10.5	15.2	21.3	23.8	25.5	24.0	22.7	17.6	8.9	1.7		
	10%	WB	-1.3	-1.8	3.8	7.8	15.0	18.1	20.0	19.4	17.9	11.6	5.2	0.1		
		MCDB	-0.6	-0.8	6.7	13.3	19.6	21.8	24.4	23.1	21.1	14.4	7.1	0.6		
(35) Mean Daily Temperature Range	MDBR		9.4	12.0	13.0	12.5	13.9	13.7	14.1	13.7	13.1	10.1	8.1	8.0		
	5% DB	MCDBR	10.3	12.5	14.9	18.9	17.5	16.3	15.7	15.2	15.4	15.0	12.8	7.6		
		MCWBR	8.8	9.6	9.4	10.2	8.1	7.8	7.4	7.7	8.2	8.9	8.7	6.3		
	5% WB	MCDBR	9.5	10.4	12.9	16.0	13.4	13.6	13.3	12.2	12.0	12.8	11.3	6.8		
		MCWBR	8.3	8.3	8.7	9.6	6.7	7.5	7.1	6.8	7.4	8.6	8.4	6.0		
(40) Clear-Sky Solar Irradiance	taub		0.237	0.264	0.305	0.327	0.359	0.360	0.361	0.357	0.339	0.306	0.280	0.249		
	taud		2.402	2.320	2.350	2.442	2.393	2.409	2.429	2.441	2.493	2.575	2.545	2.439		
	Ebn at Noon		892	922	925	932	907	905	899	890	879	869	833	835		
	Edh at Noon		68	92	105	106	115	114	110	105	91	73	61	60		
(44) All-Sky Solar Radiation	RadAvg		1.34	2.36	3.65	4.55	5.38	5.80	6.01	5.05	3.86	2.34	1.34	0.96		
	RadStd		0.12	0.26	0.34	0.46	0.45	0.45	0.29	0.31	0.23	0.27	0.16	0.11		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page